Chapter 4: PARADIGMS

4.1 Introduction

Paradigm: model or approach that **influences how people think about and design interactive systems.**

- New paradigms change how humans interact with computers.
- Each paradigm solves limitations of previous approaches

Two open questions facing every interactive-system designer

1. **Development:** How can an interactive system be developed to ensure its usability?
2. **Demonstration / Measurement:** How can the usability of an interactive system be demonstrated or measured?

4.2 Paradigms for Interaction

4.2.1 Time Sharing

Time sharing = a single computer that can support **multiple users** at the same time.

- Replaced batch processing (punched cards, paper tapes).
- Made programming interactive.
- Computer became a partner rather than just a calculator.

4.2.2 Video Display Units (VDUs)

- Information **displayed visually** instead of paper printouts.
- **Sketchpad (1962) – Ivan Sutherland**
  - Created visual models on screen.
  - Objects stored and manipulated in memory.
- Key contributions:
  - Computers can **visualize information,** not just process data.
  - Demonstrated impact of individual innovation.

4.2.3 Programming Toolkits

- Introduced ideas like word processing and mouse.
- **Toolkit: collection of tools used to build larger tools**.
- **Bootstrapping:** using **simple components** to create more complex systems.
- Small reusable components enable scalable system development.

4.2.4 Personal Computing

LOGO (Seymour Papert)

- Graphics language for children.
- Used a programmable turtle for drawing.
- Showed powerful systems can be easy to use.

Papert's Maxim: A powerful system becomes even more powerful if it is easier to use.

Alan Kay

- Promoted personal computers.
- Developed **Smalltalk**.

- Proposed **Dynabook** (portable personal computer concept).
- Goal: **computing accessible to everyone.**

4.2.5 Window Systems and the WIMP Interface

- Supports **multiple tasks simultaneously.**
- Uses separate windows for different activities.

WIMP =

- Windows
- Icons
- Menus
- Pointers
- First commercial system: **Xerox Star 8010 (1981)**.

4.2.6 The Metaphor

- **Explains new concepts** using **familiar ones.**

Problems

- May become misleading over time.
- Some computer actions have no real-world equivalent.
- Can be culturally biased.

Virtual Reality

- Extreme form of metaphor.
- User becomes part of the virtual environment.

4.2.7 Direct Manipulation

Ben Shneiderman (1982) ⇒ chp7 wali cheez!!

1. Visible objects.
2. Incremental actions.
3. Rapid feedback.
4. Reversible actions (undo).
5. Syntactically correct actions.
6. Replace commands with direct actions.

Apple Macintosh (1984)

- Popularized direct manipulation.
- Used desktop metaphor and drag-and-drop.

Model-World Metaphor

- **User acts directly on interface objects.**
- Interface becomes the user's world.

WYSIWYG - What You See Is What You Get

- Screen closely matches final output.
- Makes evaluation easier.
- Sometimes limits advanced formatting flexibility.

Worked Exercise — Word Processor & Direct Manipulation

> **Question**
>
> *Discuss the ways in which a full-page word processor is or is not a direct manipulation interface for editing a document using Shneiderman's criteria. What features of a modern word processor break the metaphor of composition with pen (or typewriter) and paper?*

Meets Shneiderman's Criteria

- Text is visible.
- Immediate feedback while typing.
- Undo supports reversibility.
- Menus/buttons ensure valid actions.
- Direct selection and dragging replace commands.

Breaks Typewriter Metaphor

- Spacebar inserts a character (active).
- Typewriter spacebar only moves cursor (passive).
- Modern word processors differ greatly from typewriters.

Passive key = a key that only moves the cursor/position and **does not create a character** in the document.

Active key = a key that **adds a character or content** to the document.

4.2.8 Language versus Action

Action Paradigm

- **User directly manipulates objects.**
- Easier for simple tasks.

Language Paradigm

- **User gives instructions.**
- Better for complex/repetitive tasks.

Two Forms

1. The user understands the system.
2. Interface/agent interprets user's intentions.

Programming by Example

- User demonstrates actions.
- System records them as reusable procedures.

Advantages & disadvantages

Aspect	Action paradigm (direct manipulation)	Language paradigm
Simple tasks	Easier; pointing/recognizing reduces identification errors	Comparatively more effort
Complex / repetitive tasks	Tedious — repeated execution with minor changes	Can describe a generic procedure once (e.g., a loop over all files) and let it run
Error risk	Lower for certain classes of error	Errors may surface late, especially if acting in the user's absence

4.2.9 Hypertext

- **Linked information** through associations.
- Non-linear linked text.

Hypermedia

- Extends linking to **text, images, audio, video, etc.**

4.2.10 Multi-modality

- Uses **multiple communication channels simultaneously.**

Modality

- A communication channel (voice, gesture, touch, etc.).

Multi-modal Systems

- Combine channels such as speech + gesture.
- Mimic natural human communication.

Each input/output device is a **communication channel** corresponding to certain human communication channels.

4.2.11 Computer-Supported Cooperative Work (CSCW)

- Supports **collaboration through computers.**

› Email

- Classic CSCW example.
- Faster than traditional mail.

Types

Asynchronous

- **Users need not be present together.**
- Example: Email.

Synchronous

- **Users communicate at same time.**
- Example: Phone call.

Groupware

- Software designed for **group collaboration**.

4.2.12 The World Wide Web

Key Components

- **HTTP, HTML, URLs**

Success Factors

- Ease of use.
- Graphical browsers.
- Reached critical mass.

Impact

- Connected people globally.
- Turned computing into a social activity.
- Enabled e-commerce and online communities.

The internet vs. the web

The Internet	The Web (built on top)
A collection of computers linked by any data connection (telephone+modem to high-bandwidth optical).	Adds its own network protocol (http).
Common transmission protocols: TCP/IP.	A standard markup notation (such as HTML) for laying out pages.

Addressing: IP addresses and domain names. Anyone can read anything anywhere (if it conforms).	A global naming scheme: URLs. Pages hold text, images, movies, sound, and hypertext links.

4.2.13 Agent-based Interfaces

Software Agents

- Perform tasks on behalf of users.

Types

Rule-Based Agents

- Follow predefined if-then rules.

Learning Agents

- Learn from user behavior.

Embodiment

- **Embodied:** visible representation (Eager's cat).
- **Diffuse:** intelligence hidden in system (Excel SUM).

Benefits

- Automate repetitive work.
- Act even when user is absent.

4.2.14 Ubiquitous Computing

where does computing happen and **where do we go** to interact? For 50 years computers were **confined to a box** on a desk/office/lab — the user knows where the computer is and must **walk over to it**.

- Computing should blend into daily life.
- Also called **Pervasive Computing**.

Vision: Technology becomes invisible.

The electric-motor analogy

- When first introduced, the electric motor was **large, loud, very noticeable**. Today households have so many we **hardly notice** them — utility led to **ubiquity and hence invisibility**.

Three scales of computing (PARC)

Scale	Size / Role	Examples
Yard	**Very large**; for wide-open public spaces; shared by a group (one per home / hallway / auditorium).	High-resolution large screens
Foot	Like **today's personal computers;** one per individual (desk, bedroom, briefcase).	Laptops, tablet computers
Inch	Truly personal, **palm-sized;** everyone owns several; as unremarkable as a pen or sticky-note pad.	phones, pagers, apple watches?

The three waves of computing

1) Many people → one computer (Mainframes)
2) One computer → one person (PCs)
3) Many devices → one person (Ubiquitous Computing)

Enabling Technologies

- Wireless networking
- Voice recognition
- Vision systems
- Pen computing
- Positioning systems

4.2.15 Sensor-based and Context-aware Interaction

Uses Sensors From

- Environment (cameras, motion sensors)
- Information world (web activity)
- Human body (heart rate, temperature)

Context-Aware Computing

- Interaction becomes **implicit**
- System predicts user needs automatically.

Examples

- Automatic doors
- Smart lighting
- Product recommendations
- Smart climate control

Principles of Appropriate Intelligence

1. **Be correct as often as possible.**
2. **Avoid serious problems when predictions are wrong.**

Significance

- One of the most radical paradigm shifts.
- Interaction may occur **without conscious user involvement**
- Represents a move toward seamless computing.